TABLE OF CONTENTS

SHEET NO.	SHEET NAME	BLOCK NAME
1	TABLE OF CONTENT	pc065d_fibv2b_toplevel
2	POWER	pc065d_fibv2b_toplevel
3	FMC_CONNECTOR	pc065d_fibv2b_toplevel
4	SFP_CONNECTORS	pc065d_fibv2b_toplevel
5	SFP_INTERFACES_0:3	pc065d_fibv2b_toplevel
6	SFP_INTERFACES_4:7	pc065d_fibv2b_toplevel
7	CLOCK	pc065d_fibv2b_toplevel
8	I2C LEVEL TRANSLATORS	pc065d_fibv2b_toplevel
9	LEVEL TRANSLATORS	pc065d_fibv2b_toplevel
10	CONNECTORS	pc065d_fibv2b_toplevel
11	EEPROM	pc065d_fibv2b_toplevel
12	page1	sfp_module
13	page1	sfp_module
14	page1	sfp_module
15	page1	sfp_module

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SHEET NO.	SHEET NAME	BLOCK NAME

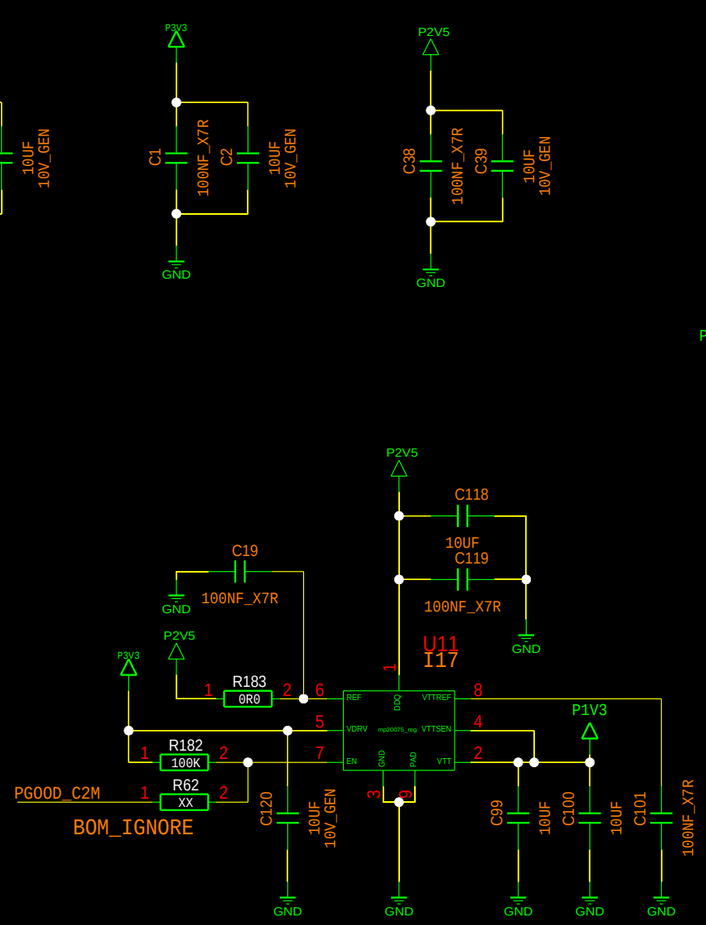
TABLE OF CONTENTS

SHEET NO.	SHEET NAME	BLOCK NAME

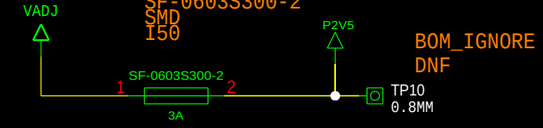
I2C ADDRESSES

EEPROM (IC1) 0X50 (ALTERNATIVE: 0X51)
TEMP MONITOR (IC2) 0X48

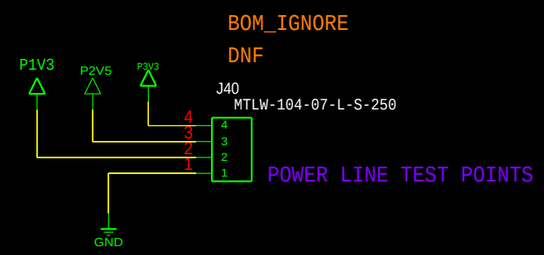
PROJECT:	pc065d_fibv2b_toplevel		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 1 / 31
	TABLE OF CONTENT		



VADJ TO DRIVE 2.5V RAIL



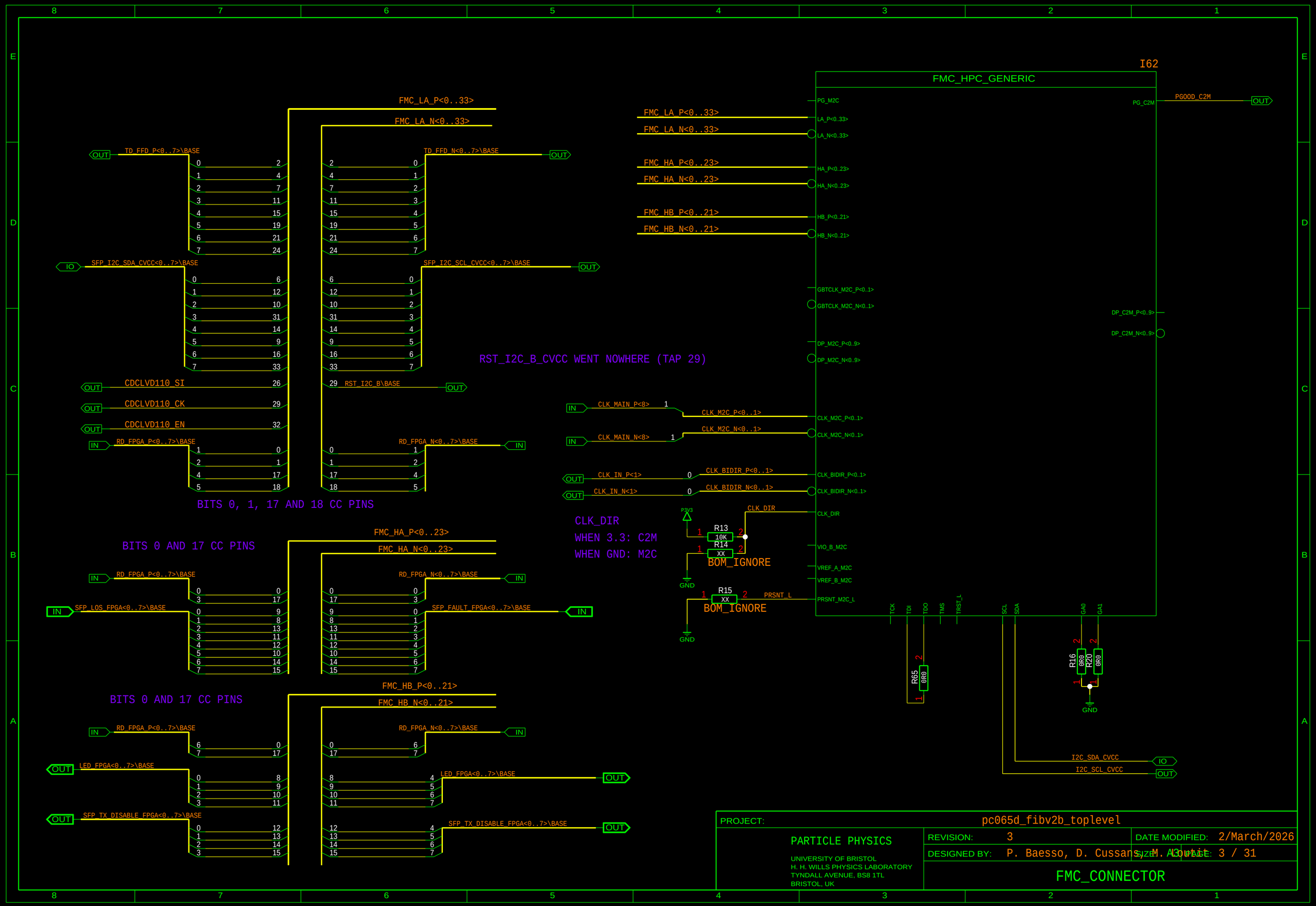
MAX CURRENTS PER RAILS
P3V3: 3 A
P12V: 1A
VADJ: 4 A



POWER LINE TEST POINTS

USE DDR REGULATOR FOR
PECL TERMINATION RAIL
(SINK CURRENT)

PROJECT:	pc065d_fibv2b_toplevel		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Alouati	SIZE: A3	SHEET: 2 / 31
	POWER		



I62
FMC_HPC_GENERIC
FMC_LA_P<0..33>
FMC_LA_N<0..33>
TD_FFD_P<0..7>\BASE
TD_FFD_N<0..7>\BASE
SFP_I2C_SDA_CVCC<0..7>\BASE
SFP_I2C_SCL_CVCC<0..7>\BASE
CDCLVD110_SI
CDCLVD110_CK
CDCLVD110_EN
RST_I2C_B\BASE
RD_FPGA_P<0..7>\BASE
RD_FPGA_N<0..7>\BASE
BITS 0, 1, 17 AND 18 CC PINS
RST_I2C_B_CVCC WENT NOWHERE (TAP 29)
CLK_MAIN_P<8>
CLK_MAIN_N<8>
CLK_M2C_P<0..1>
CLK_M2C_N<0..1>
CLK_IN_P<1>
CLK_IN_N<1>
CLK_BIDIR_P<0..1>
CLK_BIDIR_N<0..1>
CLK_DIR
WHEN 3.3: C2M
WHEN GND: M2C
R13
10K
R14
XX
BOM_IGNORE
R15
XX
BOM_IGNORE
PRSNT_L
GND
FMC_HA_P<0..23>
FMC_HA_N<0..23>
BITS 0 AND 17 CC PINS
SFP_LOS_FPGA<0..7>\BASE
SFP_FAULT_FPGA<0..7>\BASE
FMC_HB_P<0..21>
FMC_HB_N<0..21>
LED_FPGA<0..7>\BASE
SFP_TX_DISABLE_FPGA<0..7>\BASE
PG_M2C
LA_P<0..33>
LA_N<0..33>
HA_P<0..23>
HA_N<0..23>
HB_P<0..21>
HB_N<0..21>
GBTCLK_M2C_P<0..1>
GBTCLK_M2C_N<0..1>
DP_M2C_P<0..9>
DP_M2C_N<0..9>
CLK_M2C_P<0..1>
CLK_M2C_N<0..1>
CLK_BIDIR_P<0..1>
CLK_BIDIR_N<0..1>
CLK_DIR
VIO_B_M2C
VREF_A_M2C
VREF_B_M2C
PRSNT_M2C_L
PG_C2M
PGOOD_C2M
DP_C2M_P<0..9>
DP_C2M_N<0..9>
TCK
TDI
TDO
TMS
TRST_L
SCL
SDA
GA0
GA1
R65
0R0
R16
0R0
R20
0R0
I2C_SDA_CVCC
I2C_SCL_CVCC
PROJECT: pc065d_fibv2b_toplevel
PARTICLE PHYSICS
UNIVERSITY OF BRISTOL
H. H. WILLS PHYSICS LABORATORY
TYNDALL AVENUE, BS8 1TL
BRISTOL, UK
REVISION: 3
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DESIGNED BY: P. Baesso, D. Cussans, M. Loutit
SIZE: A3
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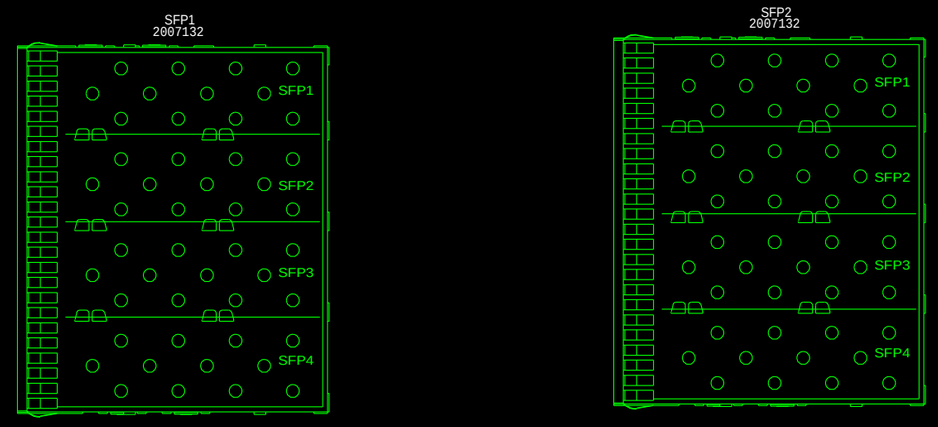
FMC_CONNECTOR

1X4 SFP+ CAGE
USE 2110069-1 FOR LIGHTPIPES
USE 2007132-1 FOR NO LIGHTGUIDES



PROJECT:	pc065d_fibv2b_toplevel		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
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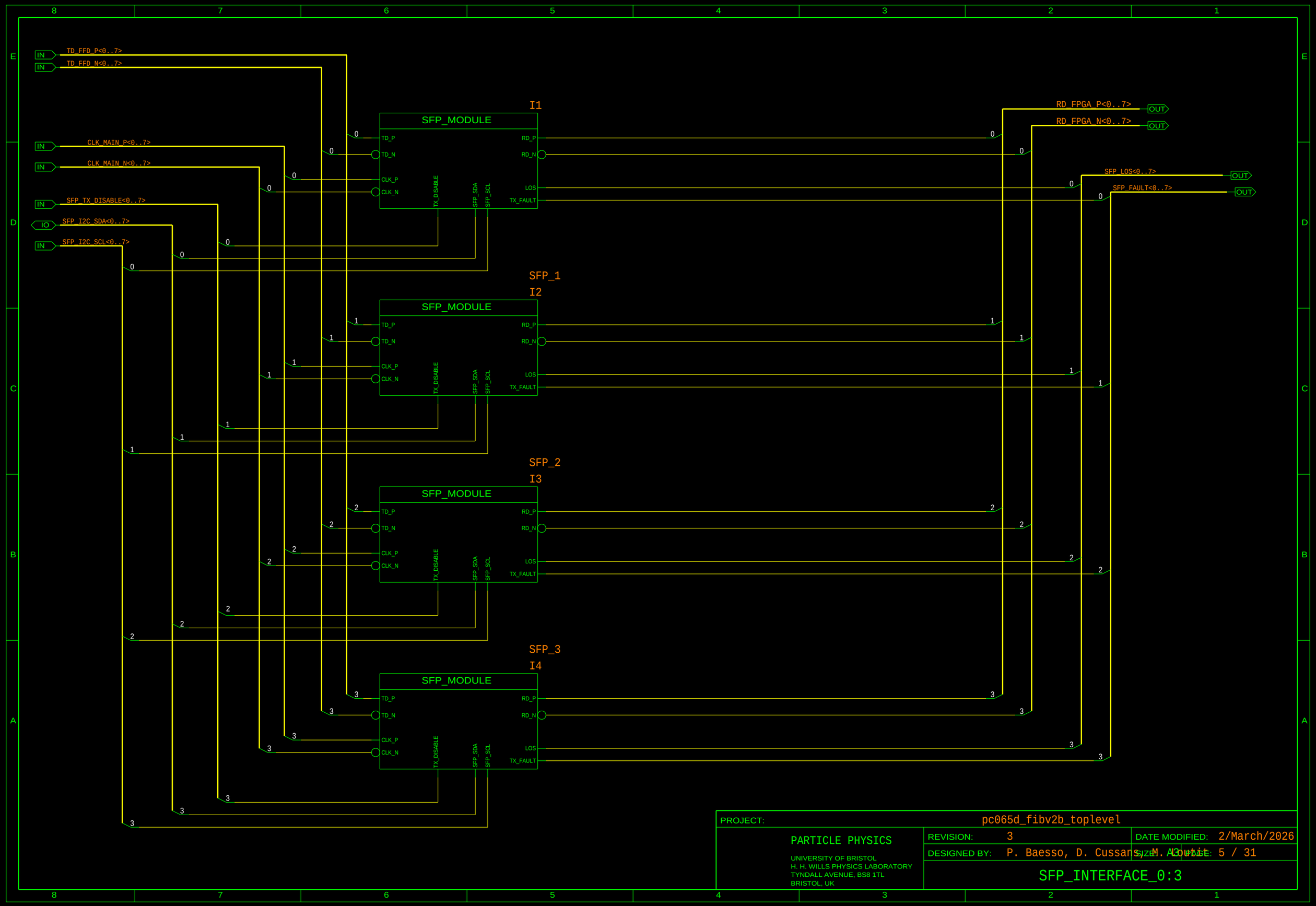
	SFP_CONNECTORS		



TD_FFD_P<0..7>
TD_FFD_N<0..7>
CLK_MAIN_P<0..7>
CLK_MAIN_N<0..7>
SFP_TX_DISABLE<0..7>
SFP_I2C_SDA<0..7>
SFP_I2C_SCL<0..7>
IN
IO
I1
SFP_MODULE
SFP_1
I2
SFP_2
I3
SFP_3
I4
TD_P
TD_N
CLK_P
CLK_N
TX_DISABLE
SFP_SDA
SFP_SCL
RD_P
RD_N
LOS
TX_FAULT
RD_FPGA_P<0..7>
RD_FPGA_N<0..7>
SFP_LOS<0..7>
SFP_FAULT<0..7>
OUT
PROJECT: pc065d_fibv2b_toplevel
PARTICLE PHYSICS
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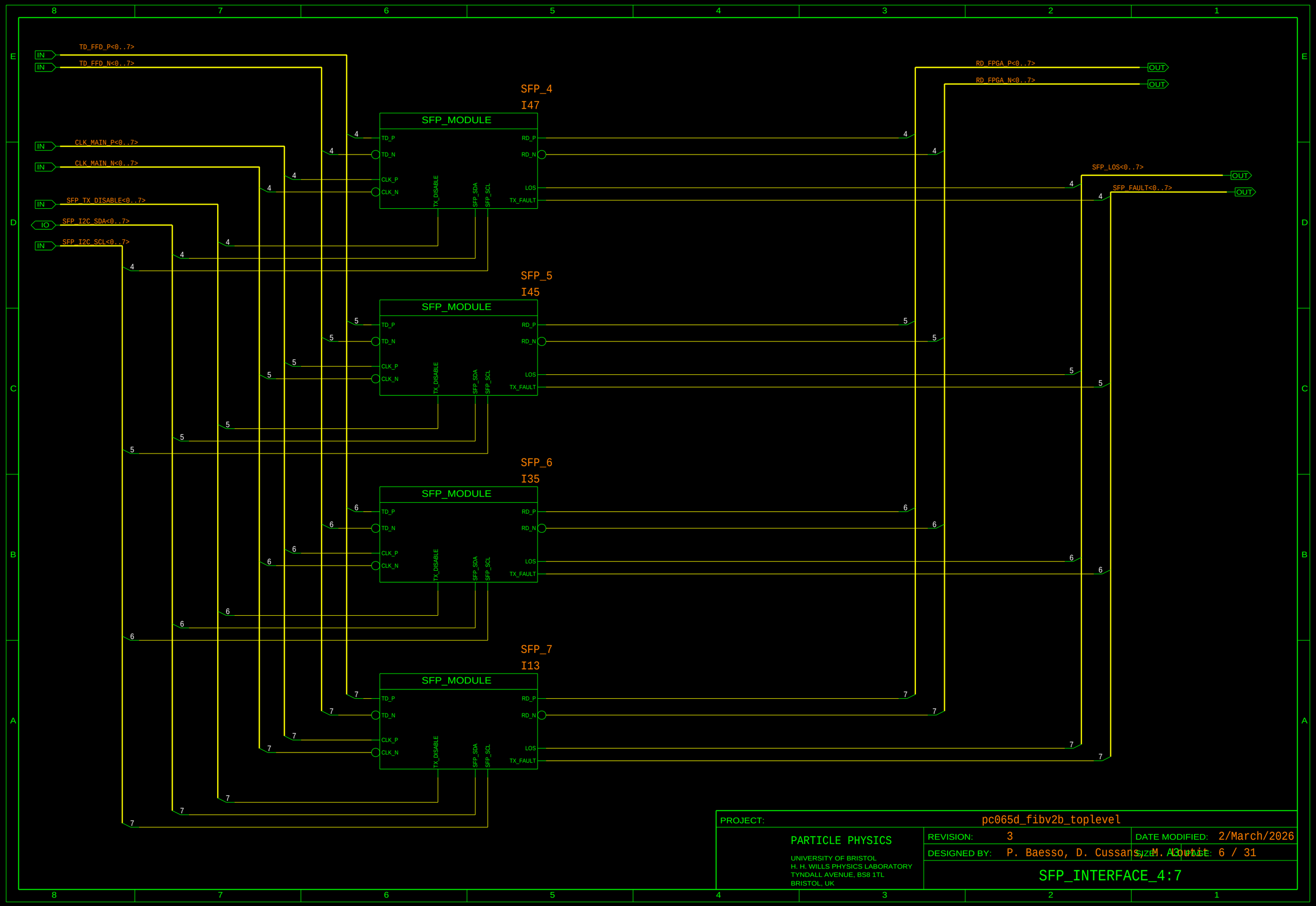
SFP_INTERFACE_0:3



TD_FFD_P<0..7>
TD_FFD_N<0..7>
CLK_MAIN_P<0..7>
CLK_MAIN_N<0..7>
SFP_TX_DISABLE<0..7>
SFP_I2C_SDA<0..7>
SFP_I2C_SCL<0..7>
RD_FPGA_P<0..7>
RD_FPGA_N<0..7>
SFP_LOS<0..7>
SFP_FAULT<0..7>
IN
IO
OUT
SFP_4
I47
SFP_5
I45
SFP_6
I35
SFP_7
I13
SFP_MODULE
TD_P
TD_N
CLK_P
CLK_N
TX_DISABLE
SFP_SDA
SFP_SCL
RD_P
RD_N
LOS
TX_FAULT
PROJECT:
pc065d_fibv2b_toplevel
PARTICLE PHYSICS
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TYNDALL AVENUE, BS8 1TL
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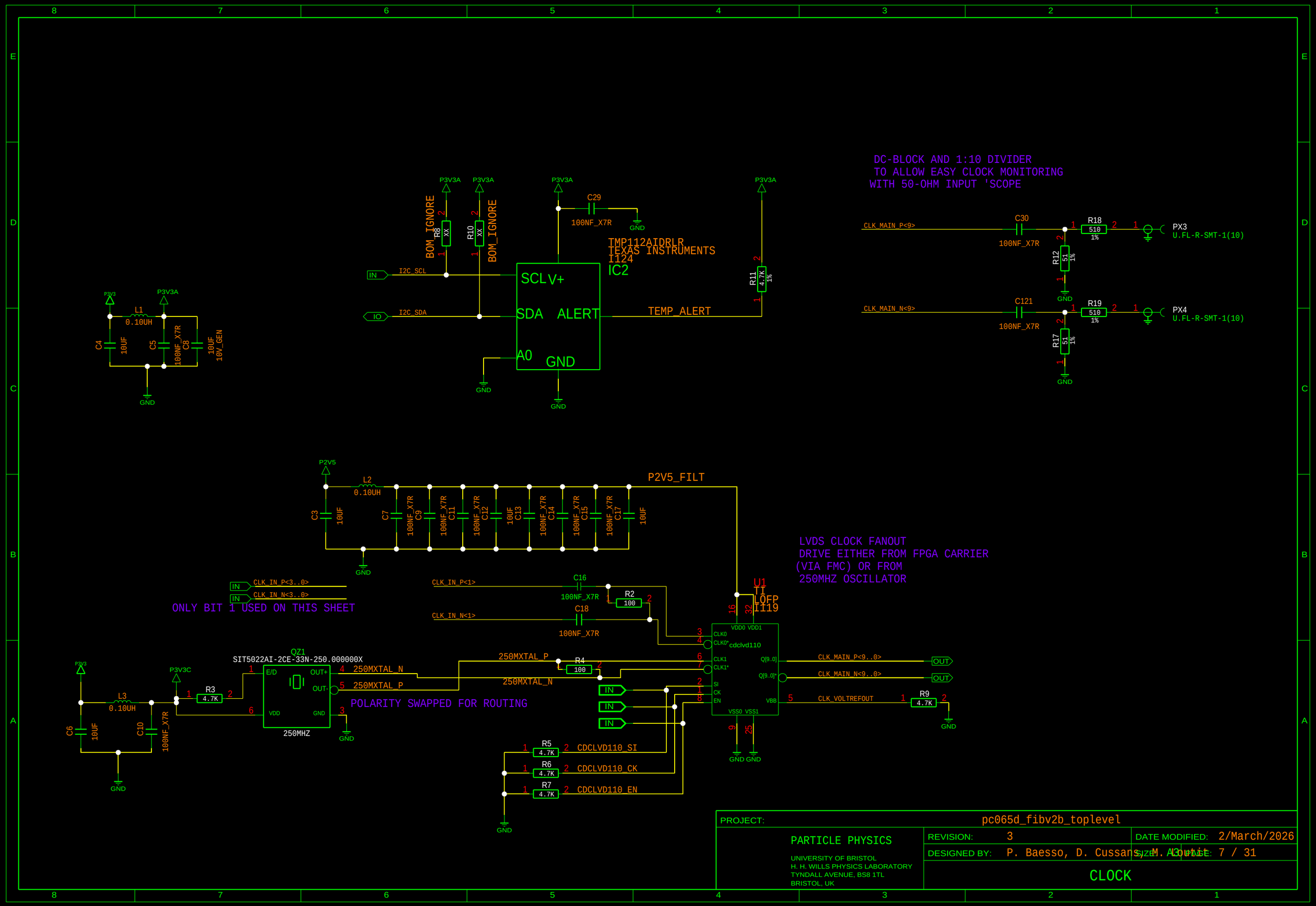
SFP_INTERFACE_4:7


DC-BLOCK AND 1:10 DIVIDER
TO ALLOW EASY CLOCK MONITORING
WITH 50-OHM INPUT 'SCOPE
TMP112AIDRLR
TEXAS INSTRUMENTS
I124
IC2
SCL
V+
SDA
ALERT
A0
GND
TEMP_ALERT
BOM_IGNORE
I2C_SCL
I2C_SDA
CLK_MAIN_P<9>
CLK_MAIN_N<9>
PX3
U.FL-R-SMT-1(10)
PX4
U.FL-R-SMT-1(10)
P2V5_FILT
LVDS CLOCK FANOUT
DRIVE EITHER FROM FPGA CARRIER
(VIA FMC) OR FROM
250MHZ OSCILLATOR
U1
TI
LOFP
I119
cdclvd110
CLK_IN_P<3..0>
CLK_IN_N<3..0>
ONLY BIT 1 USED ON THIS SHEET
CLK_IN_P<1>
CLK_IN_N<1>
QZ1
SIT5022AI-2CE-33N-250.000000X
250MHZ
250MXTAL_N
250MXTAL_P
POLARITY SWAPPED FOR ROUTING
CLK_MAIN_P<9..0>
CLK_MAIN_N<9..0>
CLK_VOLTREFOUT
CDCLVD110_SI
CDCLVD110_CK
CDCLVD110_EN
PROJECT: pc065d_fibv2b_toplevel
PARTICLE PHYSICS
UNIVERSITY OF BRISTOL
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TYNDALL AVENUE, BS8 1TL
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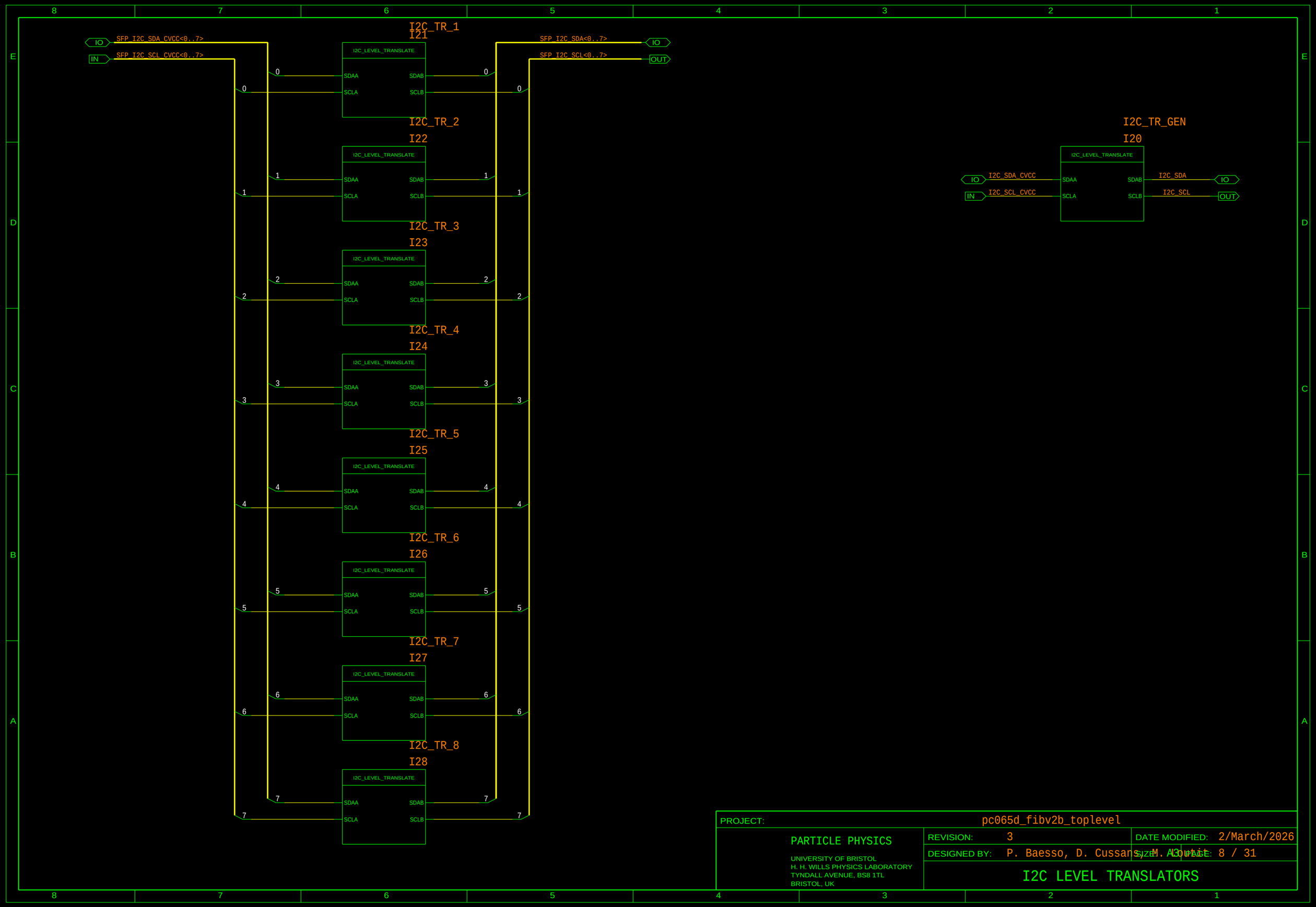
CLOCK



I2C_TR_1
I21
I2C_TR_2
I22
I2C_TR_3
I23
I2C_TR_4
I24
I2C_TR_5
I25
I2C_TR_6
I26
I2C_TR_7
I27
I2C_TR_8
I28
I2C_TR_GEN
I20
I2C_LEVEL_TRANSLATE
SDAA
SCLA
SDAB
SCLB
IO
IN
OUT
SFP_I2C_SDA_CVCC<0..7>
SFP_I2C_SCL_CVCC<0..7>
SFP_I2C_SDA<0..7>
SFP_I2C_SCL<0..7>
I2C_SDA_CVCC
I2C_SCL_CVCC
I2C_SDA
I2C_SCL
PROJECT: pc065d_fibv2b_toplevel
PARTICLE PHYSICS
UNIVERSITY OF BRISTOL
H. H. WILLS PHYSICS LABORATORY
TYNDALL AVENUE, BS8 1TL
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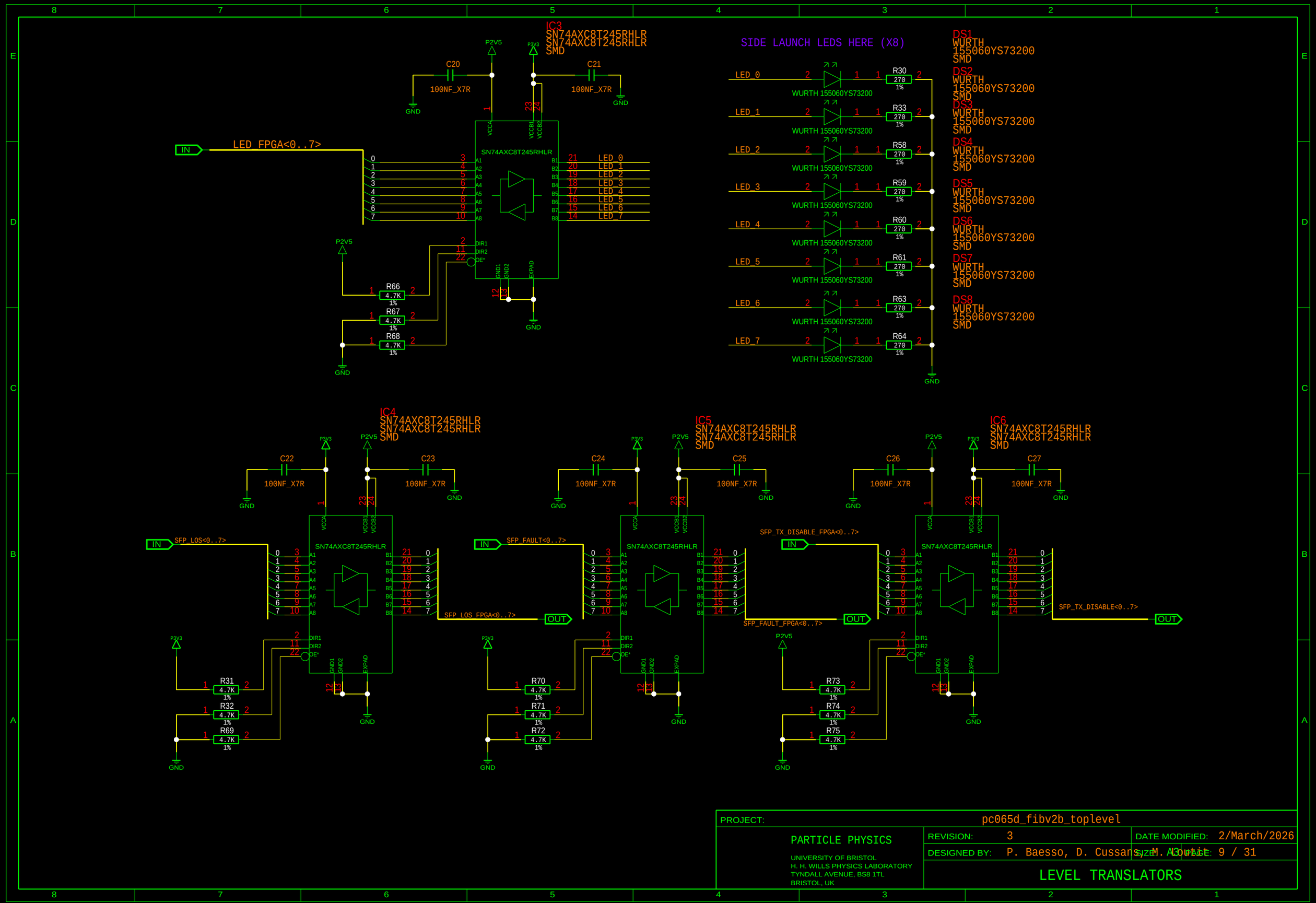
I2C LEVEL TRANSLATORS


IC3
SN74AXC8T245RHLR
SN74AXC8T245RHLR
SMD
SIDE LAUNCH LEDS HERE (X8)
DS1
WURTH
155060YS73200
SMD
DS2
WURTH
155060YS73200
SMD
DS3
WURTH
155060YS73200
SMD
DS4
WURTH
155060YS73200
SMD
DS5
WURTH
155060YS73200
SMD
DS6
WURTH
155060YS73200
SMD
DS7
WURTH
155060YS73200
SMD
DS8
WURTH
155060YS73200
SMD
LED_FPGA<0..7>
IN
SN74AXC8T245RHLR
LED_0
LED_1
LED_2
LED_3
LED_4
LED_5
LED_6
LED_7
R30
R33
R58
R59
R60
R61
R63
R64
270
1%
WURTH 155060YS73200
C20
C21
100NF_X7R
P2V5
P3V3
GND
R66
R67
R68
4.7K
IC4
SN74AXC8T245RHLR
SN74AXC8T245RHLR
SMD
C22
C23
SFP_LOS<0..7>
SFP_LOS_FPGA<0..7>
OUT
R31
R32
R69
IC5
SN74AXC8T245RHLR
SN74AXC8T245RHLR
SMD
C24
C25
SFP_FAULT<0..7>
SFP_FAULT_FPGA<0..7>
R70
R71
R72
IC6
SN74AXC8T245RHLR
SN74AXC8T245RHLR
SMD
C26
C27
SFP_TX_DISABLE_FPGA<0..7>
SFP_TX_DISABLE<0..7>
R73
R74
R75
PROJECT: pc065d_fibv2b_toplevel
PARTICLE PHYSICS
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H. H. WILLS PHYSICS LABORATORY
TYNDALL AVENUE, BS8 1TL
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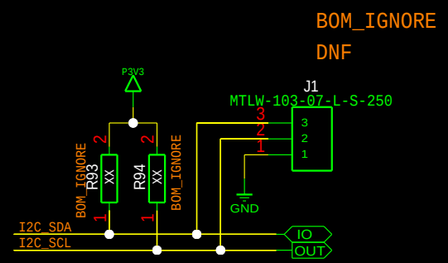
LEVEL TRANSLATORS



PROJECT:	pc065d_fibv2b_toplevel		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
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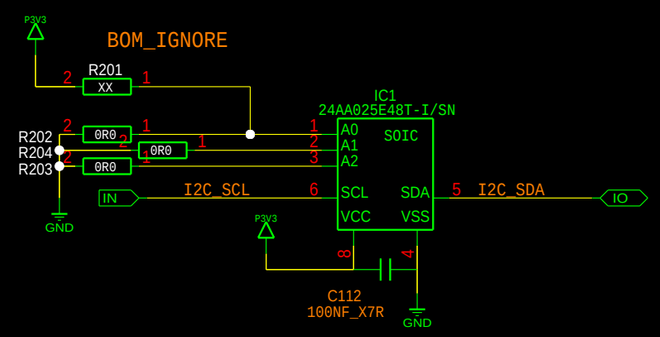
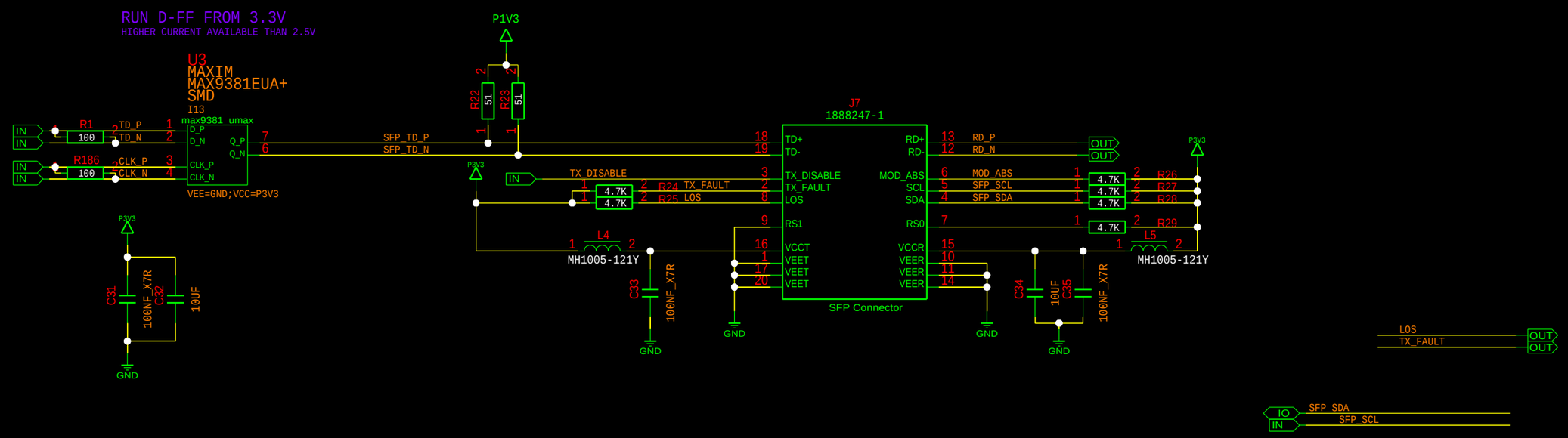
	CONNECTORS		

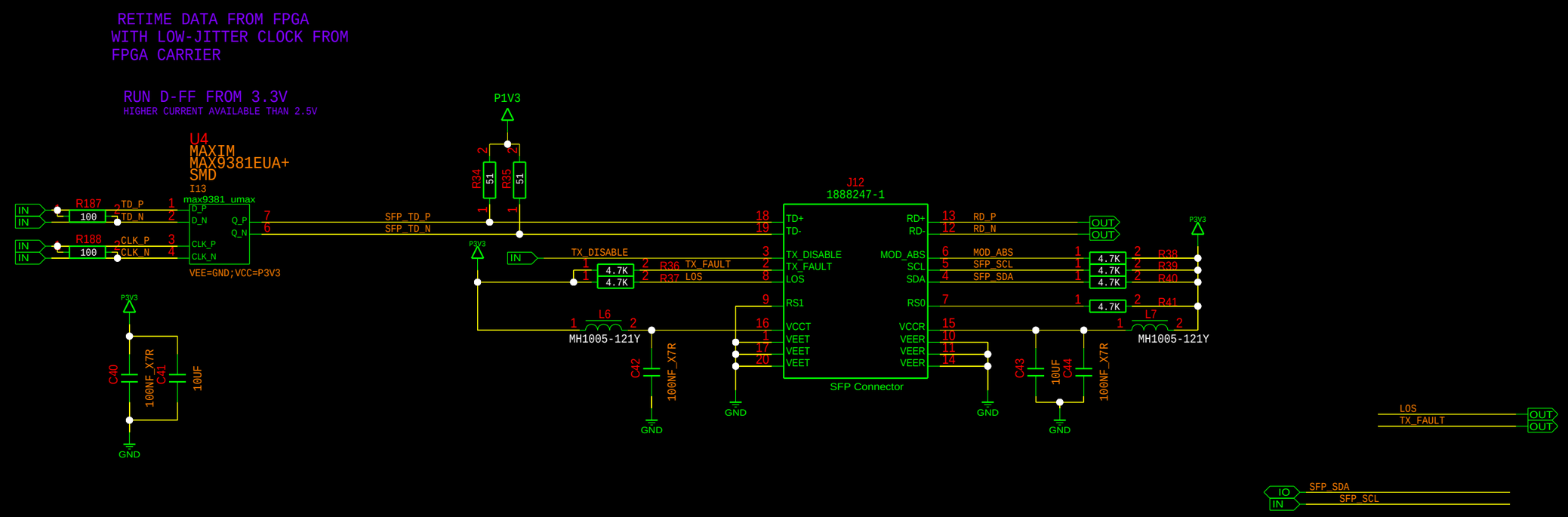
I2C EEPROM STORES PARAMETERS
AND UNIQUE ID



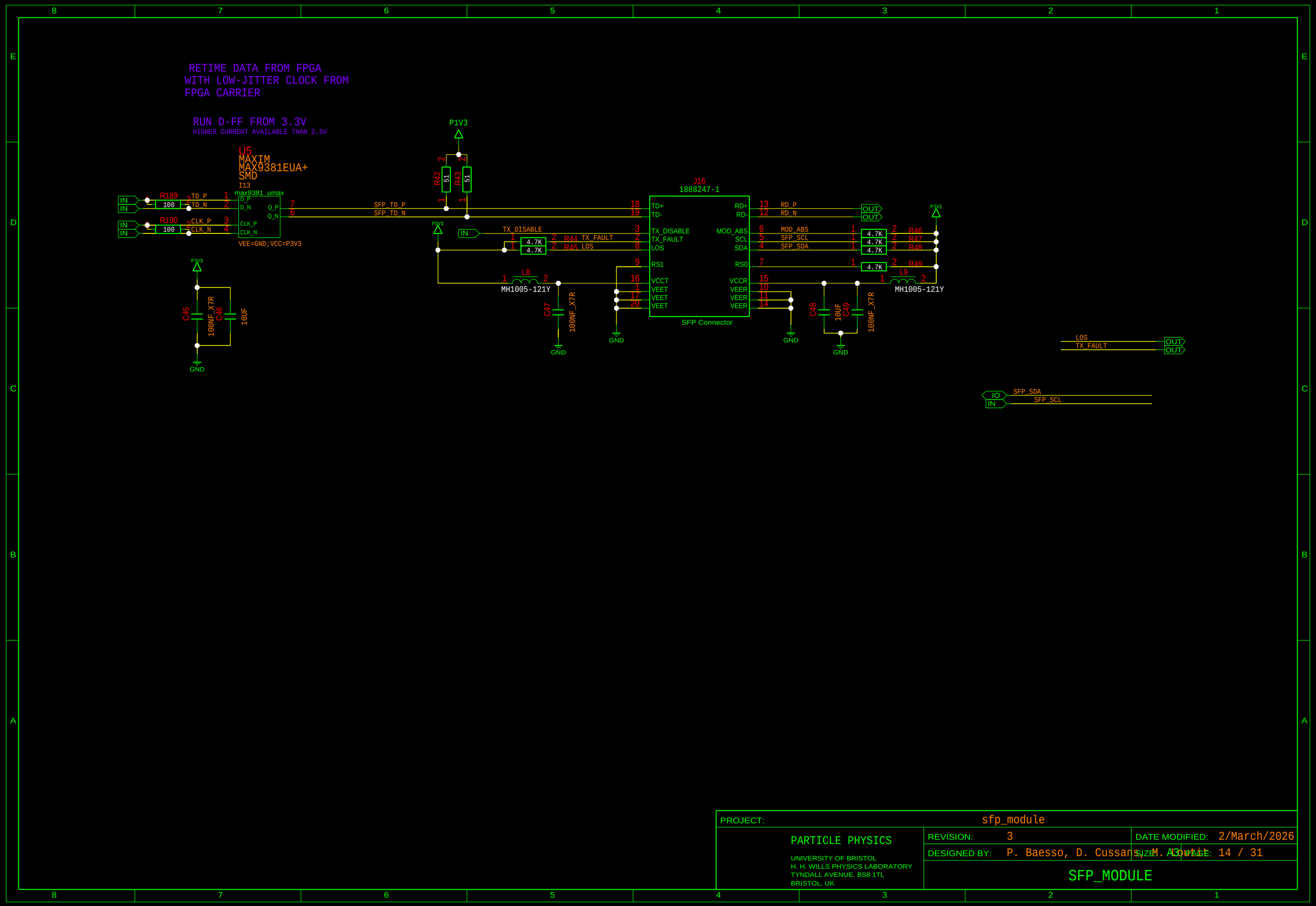
PROJECT:	pc065d_fibv2b_toplevel		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 11 / 31
	EEPROM		



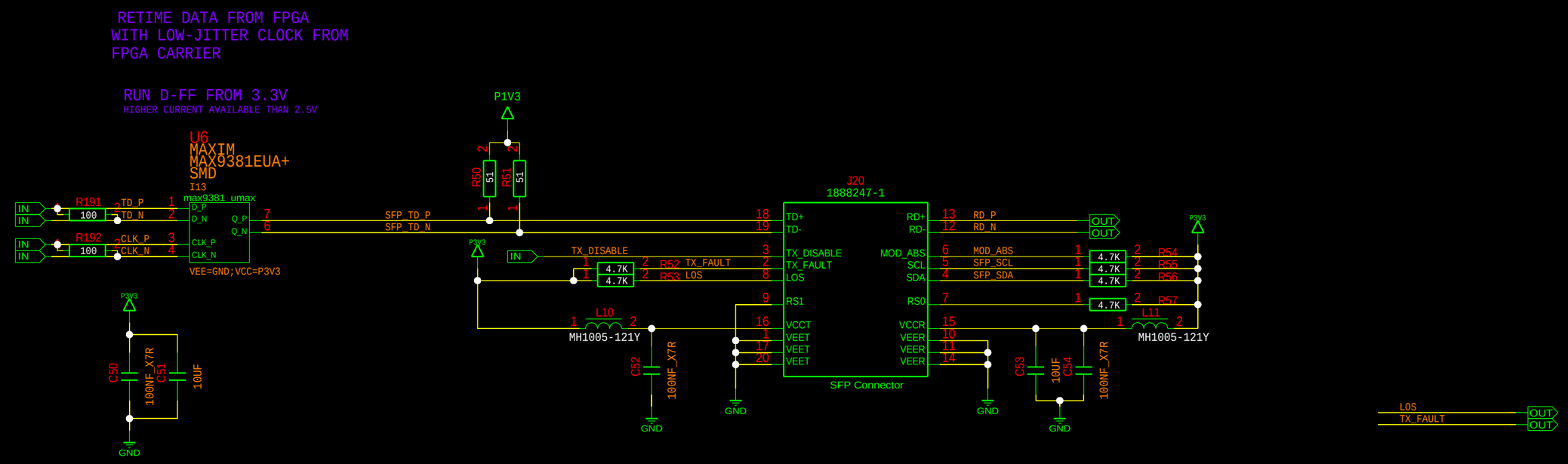
PROJECT:	sfp_module		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Alouati	SIZE: A3	SHEET: 12 / 31
	SFP_MODULE		



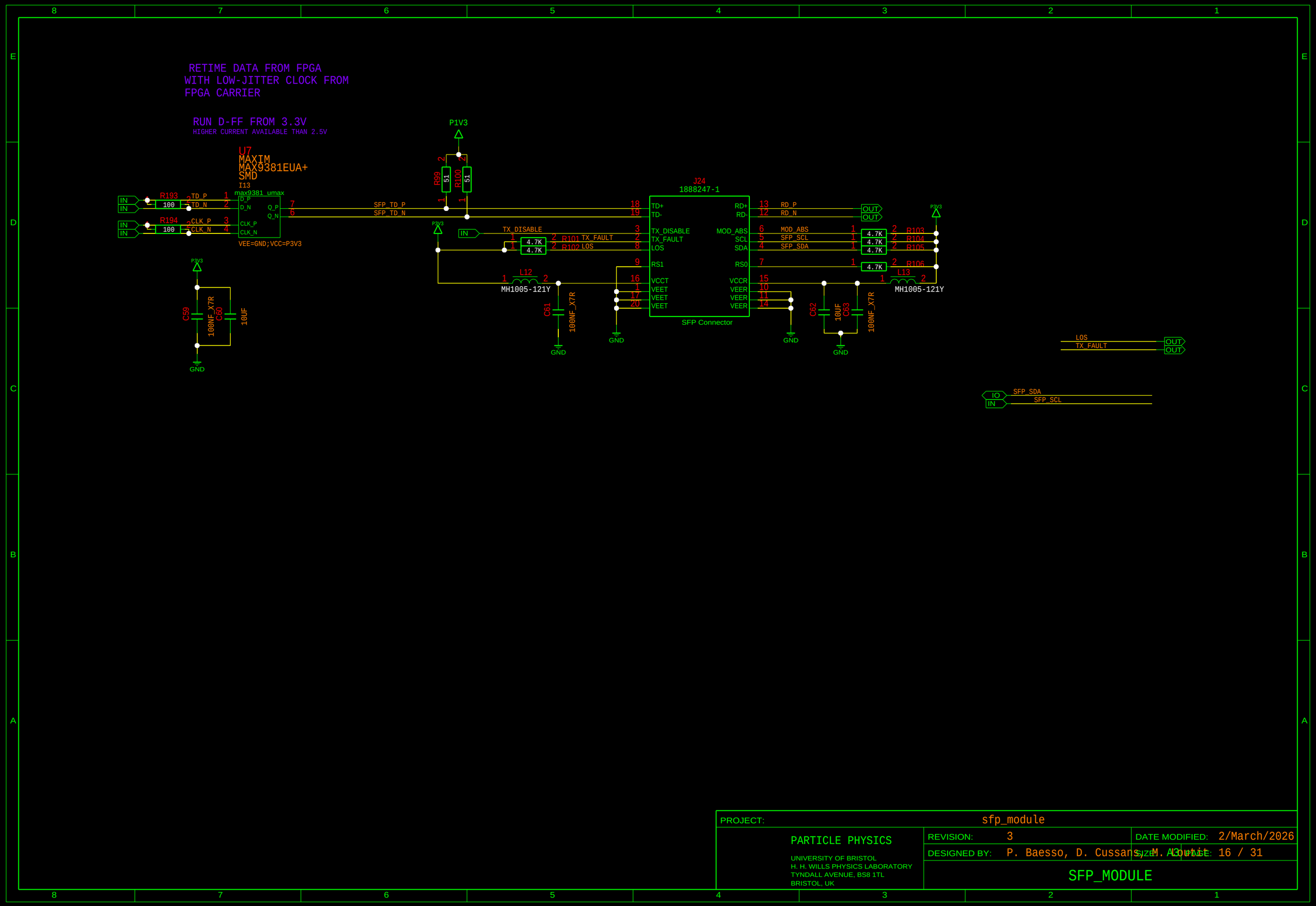
PROJECT:	sfp_module		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Aoulit	SIZE: A3	PAGE: 13 / 31
	SFP_MODULE		



PROJECT:	sfp_module		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 14 / 31
	SFP_MODULE		



PROJECT:	sfp_module		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 15 / 31
	SFP_MODULE		


RETIME DATA FROM FPGA
WITH LOW-JITTER CLOCK FROM
FPGA CARRIER
RUN D-FF FROM 3.3V
HIGHER CURRENT AVAILABLE THAN 2.5V
U7
MAXIM
MAX9381EUA+
SMD
I13
max9381_umax
VEE=GND;VCC=P3V3
P1V3
R99
R100
J24
1888247-1
SFP Connector
TX_DISABLE
TX_FAULT
LOS
RS1
VCCT
VEET
MOD_ABS
SCL
SDA
RS0
VCCR
VEER
RD+
RD-
TD+
TD-
SFP_TD_P
SFP_TD_N
RD_P
RD_N
MOD_ABS
SFP_SCL
SFP_SDA
R101
R102
R103
R104
R105
R106
L12
L13
MH1005-121Y
C59
C60
C61
C62
C63
100NF_X7R
10UF
GND
IN
OUT
IO
SFP_SDA
SFP_SCL
PROJECT:
sfp_module
PARTICLE PHYSICS
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TYNDALL AVENUE, BS8 1TL
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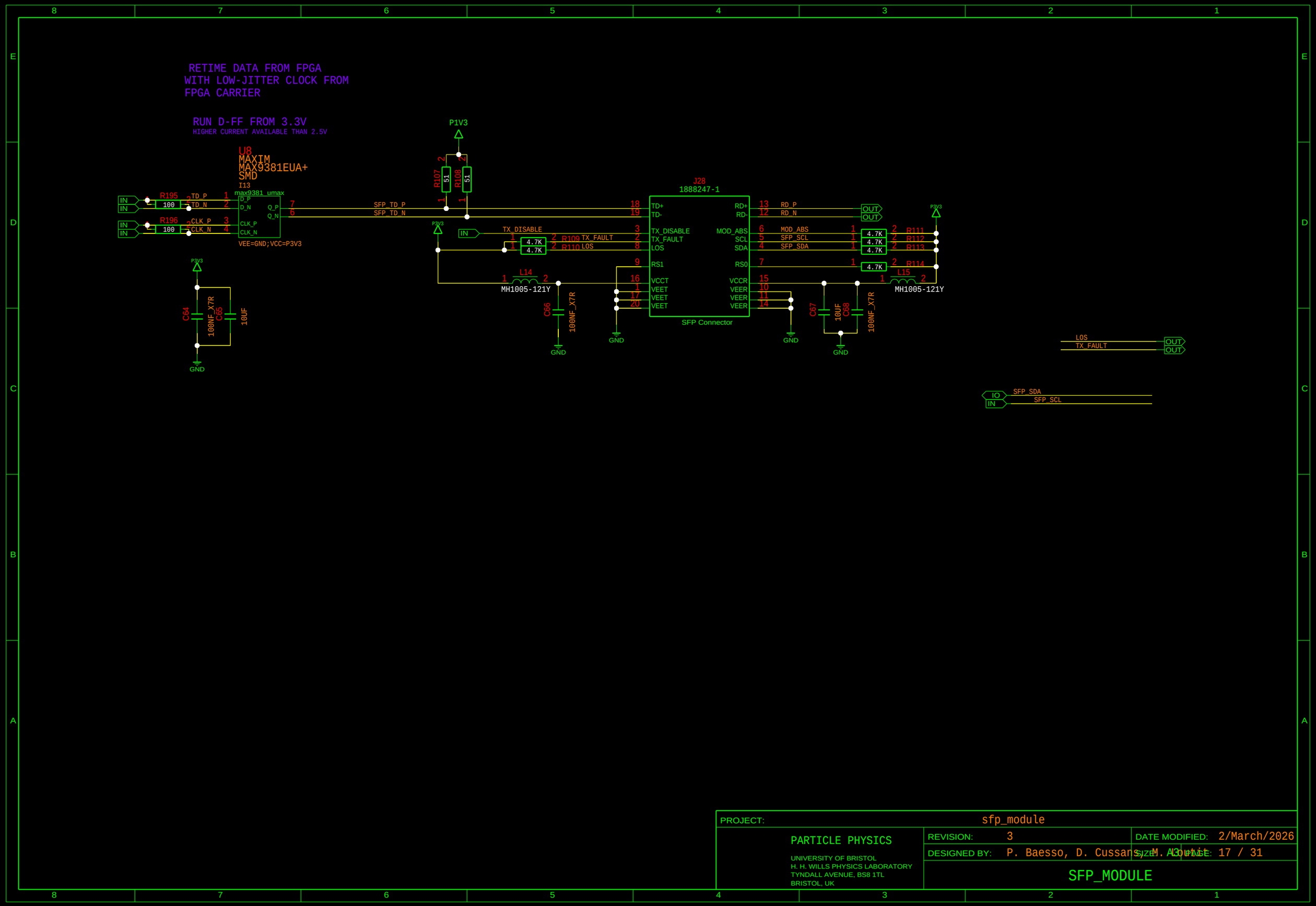
SFP_MODULE


RETIME DATA FROM FPGA
WITH LOW-JITTER CLOCK FROM
FPGA CARRIER
RUN D-FF FROM 3.3V
HIGHER CURRENT AVAILABLE THAN 2.5V
U8
MAXIM
MAX9381EUA+
SMD
I13
max9381_umax
VEE=GND;VCC=P3V3
P1V3
J28
1888247-1
SFP Connector
TX_DISABLE
TX_FAULT
LOS
RS1
VCCT
VEET
RD+
RD-
MOD_ABS
SCL
SDA
RS0
VCCR
VEER
SFP_TD_P
SFP_TD_N
RD_P
RD_N
SFP_SCL
SFP_SDA
MH1005-121Y
100NF_X7R
10UF
GND
IN
OUT
IO
PROJECT:
sfp_module
PARTICLE PHYSICS
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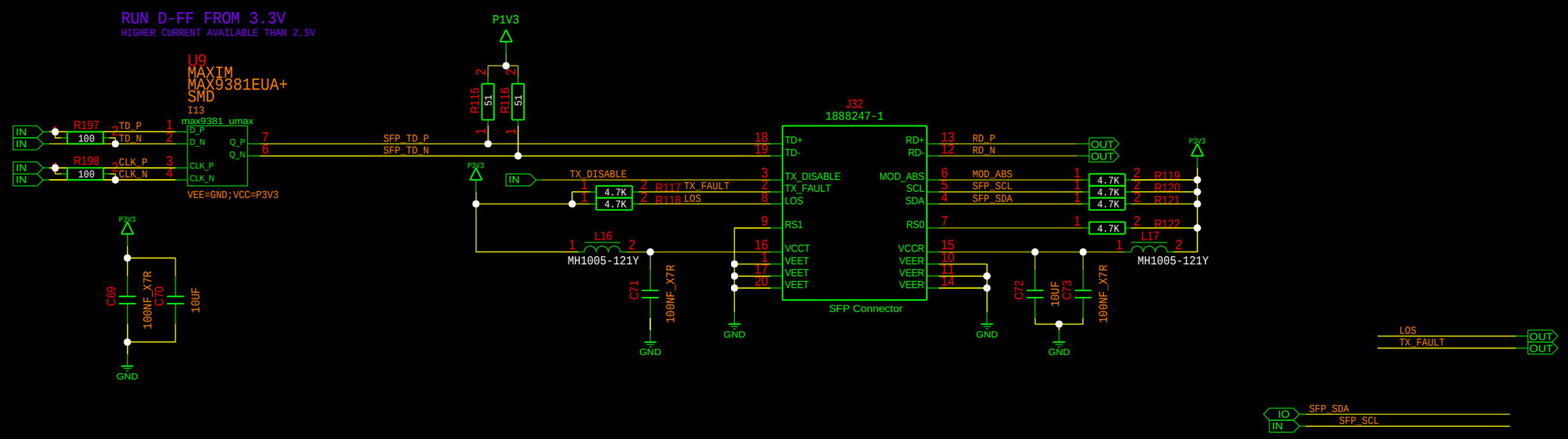
TYNDALL AVENUE, BS8 1TL
BRISTOL, UK
REVISION: 3
DATE MODIFIED: 2/March/2026
DESIGNED BY: P. Baesso, D. Cussans, M. Aoulit
SIZE: A3
PAGE: 17 / 31
SFP_MODULE



PROJECT:	sfp_module		
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Alouati	SIZE: A3	SHEET: 18 / 31
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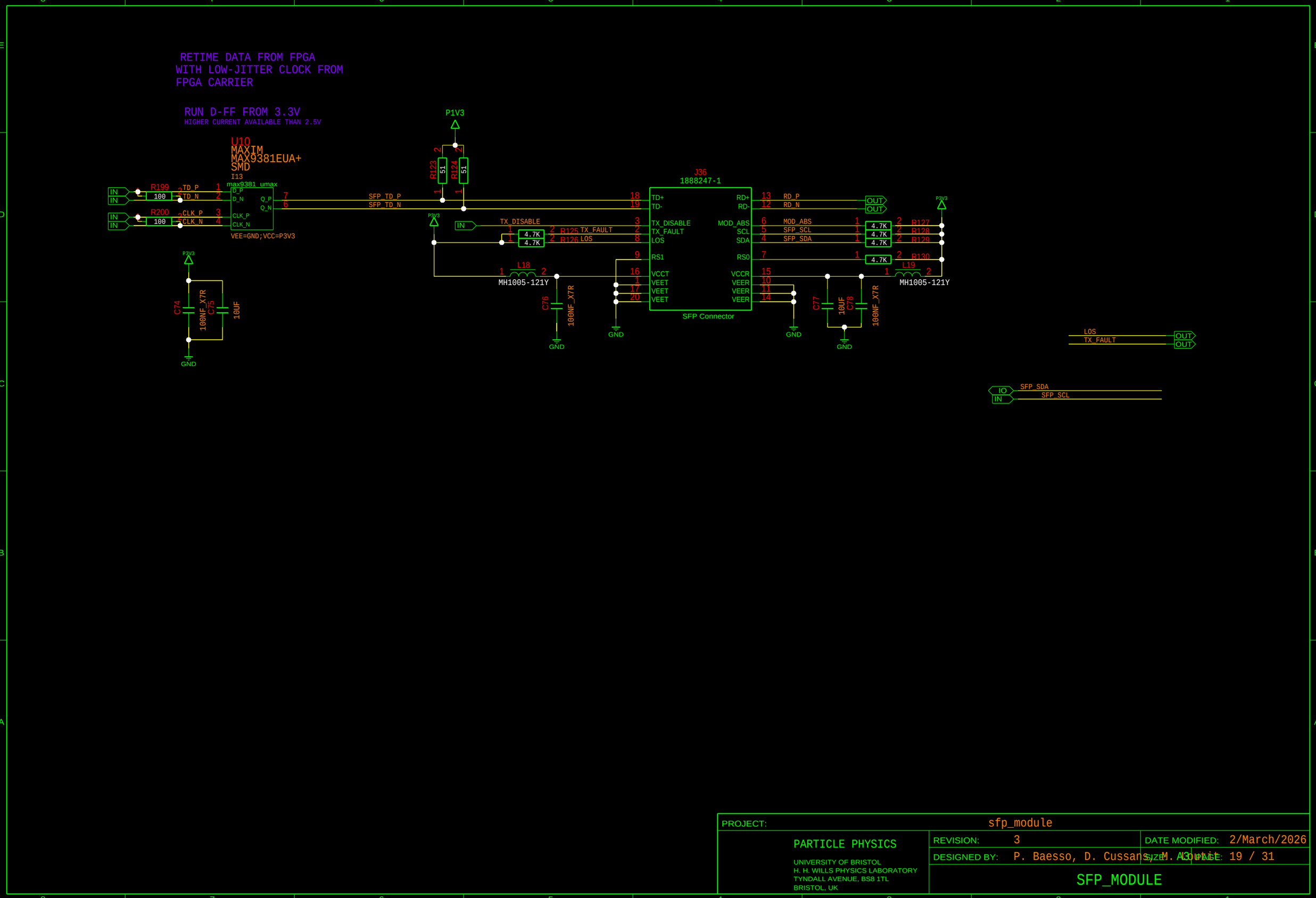

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PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 19 / 31
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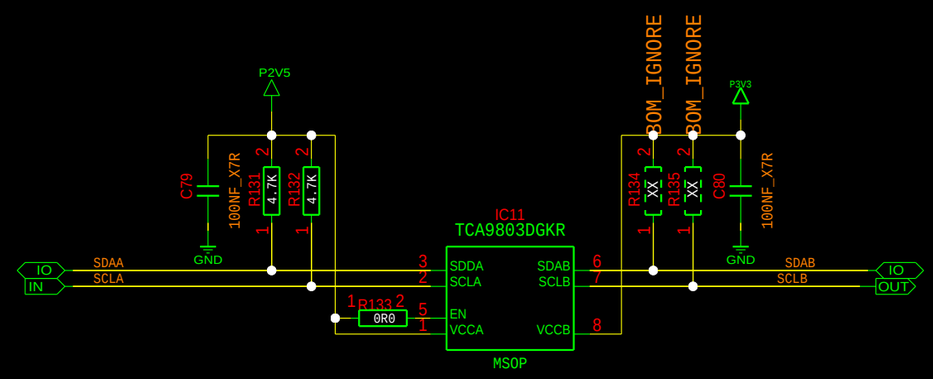

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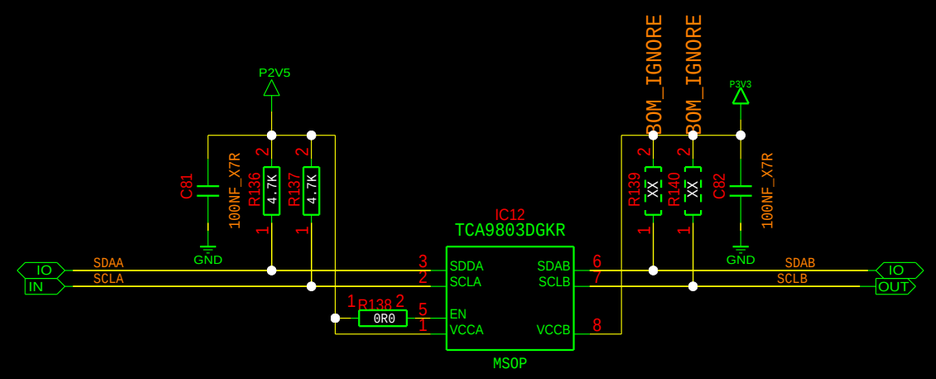
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 20 / 31
	I2C LEVEL TRANSLATOR		



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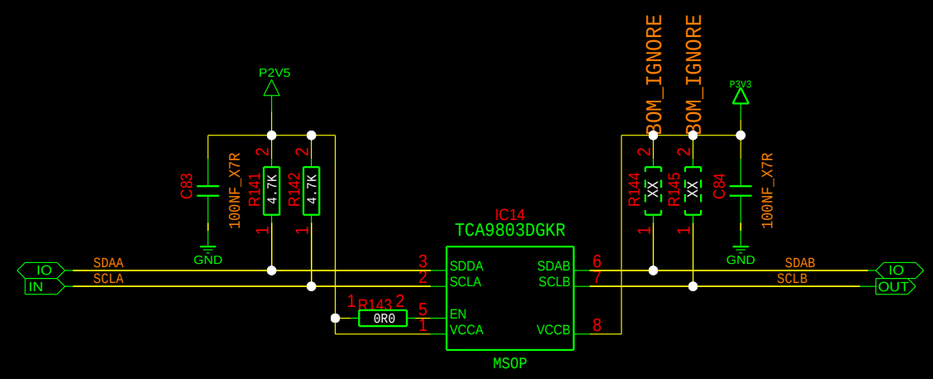
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 21 / 31
	I2C LEVEL TRANSLATOR		



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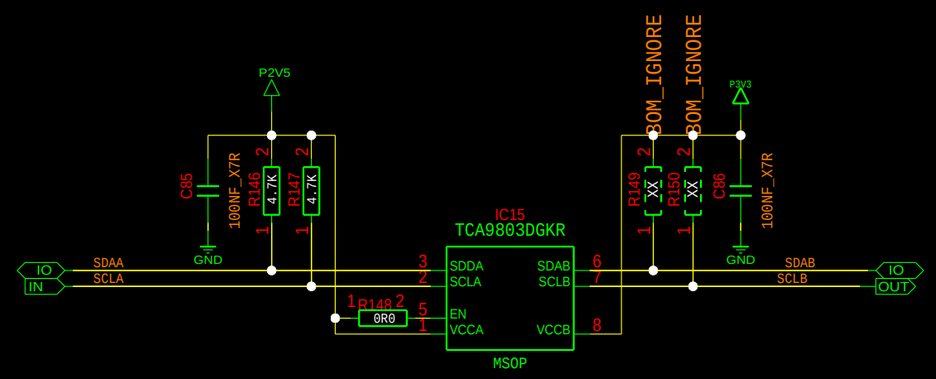
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 22 / 31
	I2C LEVEL TRANSLATOR		



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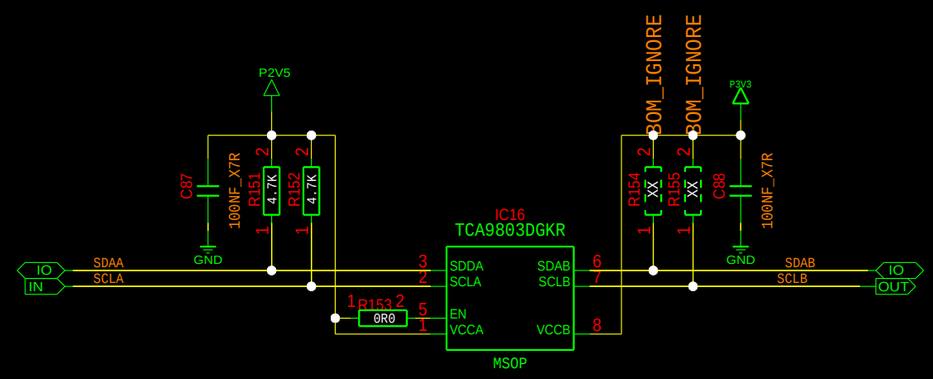
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 23 / 31
	I2C LEVEL TRANSLATOR		



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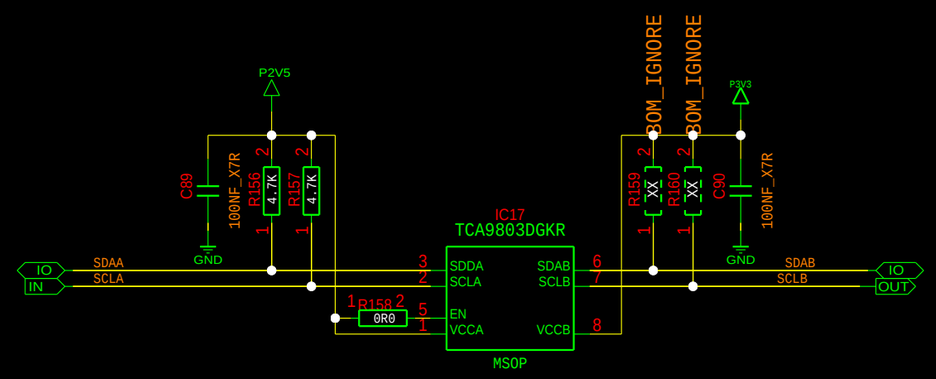
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 24 / 31
	I2C LEVEL TRANSLATOR		



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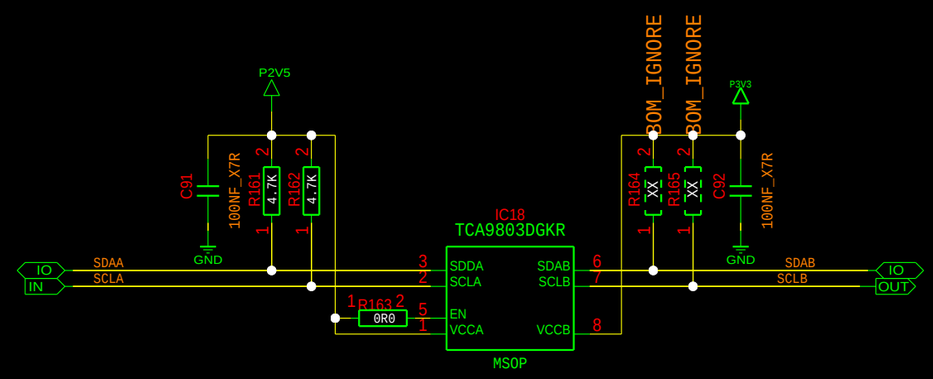
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 25 / 31
	I2C LEVEL TRANSLATOR		



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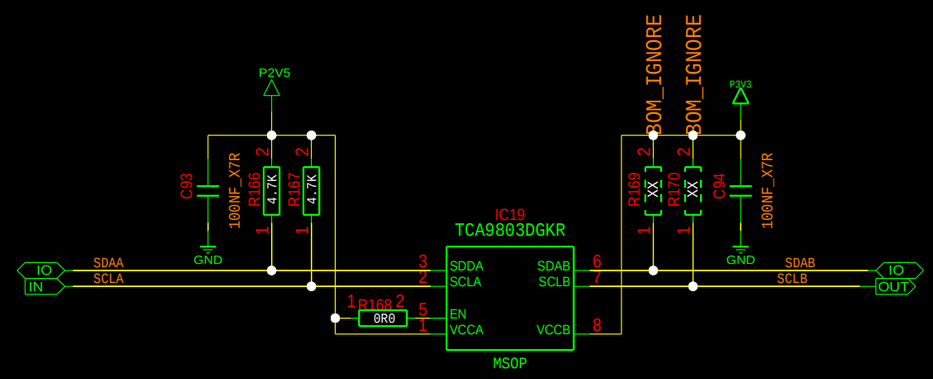
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 26 / 31
	I2C LEVEL TRANSLATOR		



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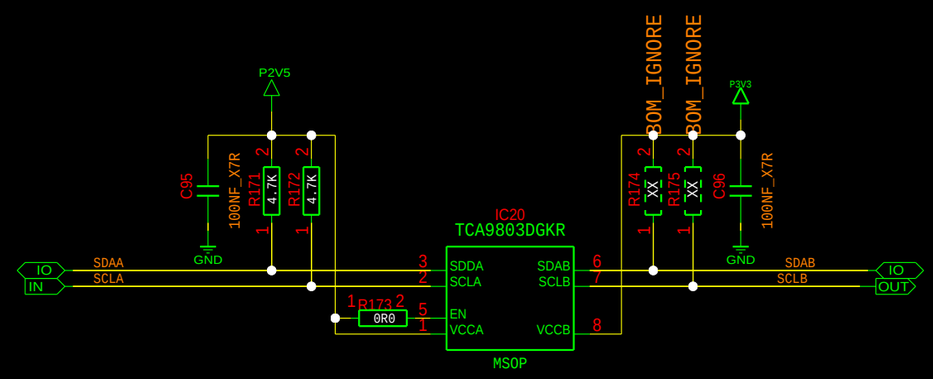
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 27 / 31
	I2C LEVEL TRANSLATOR		



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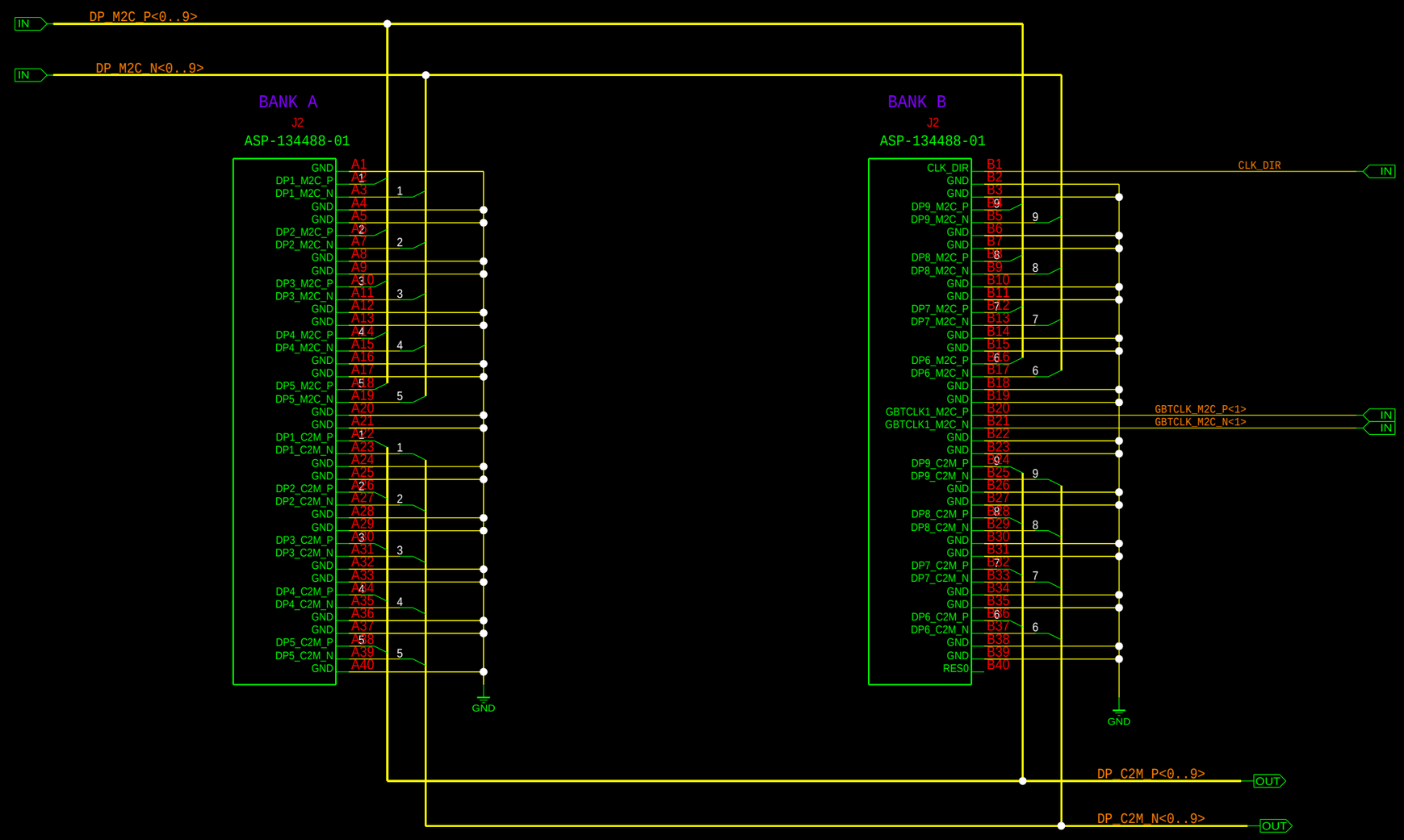
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 28 / 31
	I2C LEVEL TRANSLATOR		



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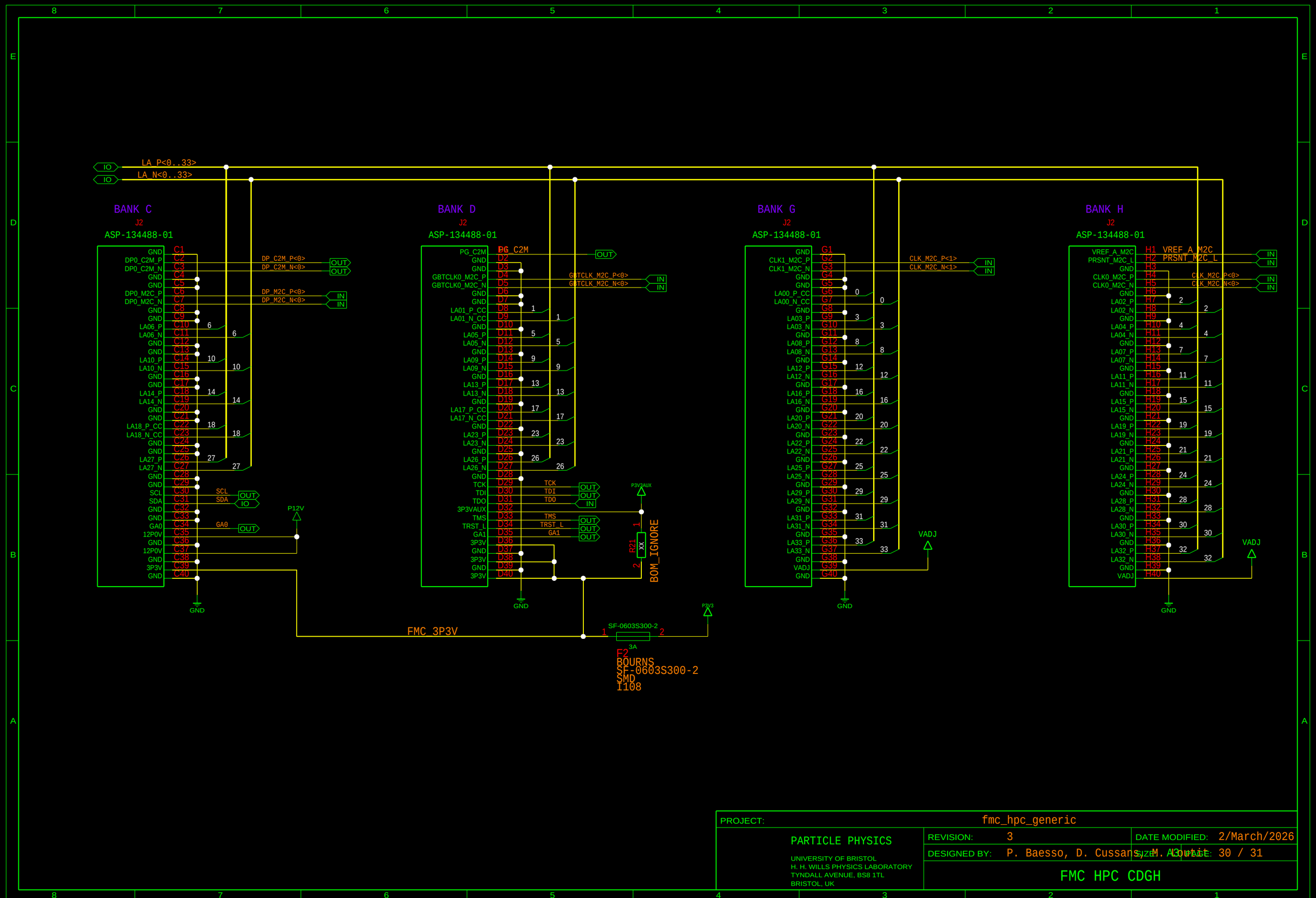
PARTICLE PHYSICS UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	REVISION: 3	DATE MODIFIED: 2/March/2026	
	DESIGNED BY: P. Baesso, D. Cussans, M. Alouche	SHEET: 29 / 31	
	FMC HPC AB		



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PARTICLE PHYSICS	REVISION: 3	DATE MODIFIED: 2/March/2026	
UNIVERSITY OF BRISTOL H. H. WILLS PHYSICS LABORATORY TYNDALL AVENUE, BS8 1TL BRISTOL, UK	DESIGNED BY: P. Baesso, D. Cussans, M. Loutit	SIZE: A3	PAGE: 30 / 31
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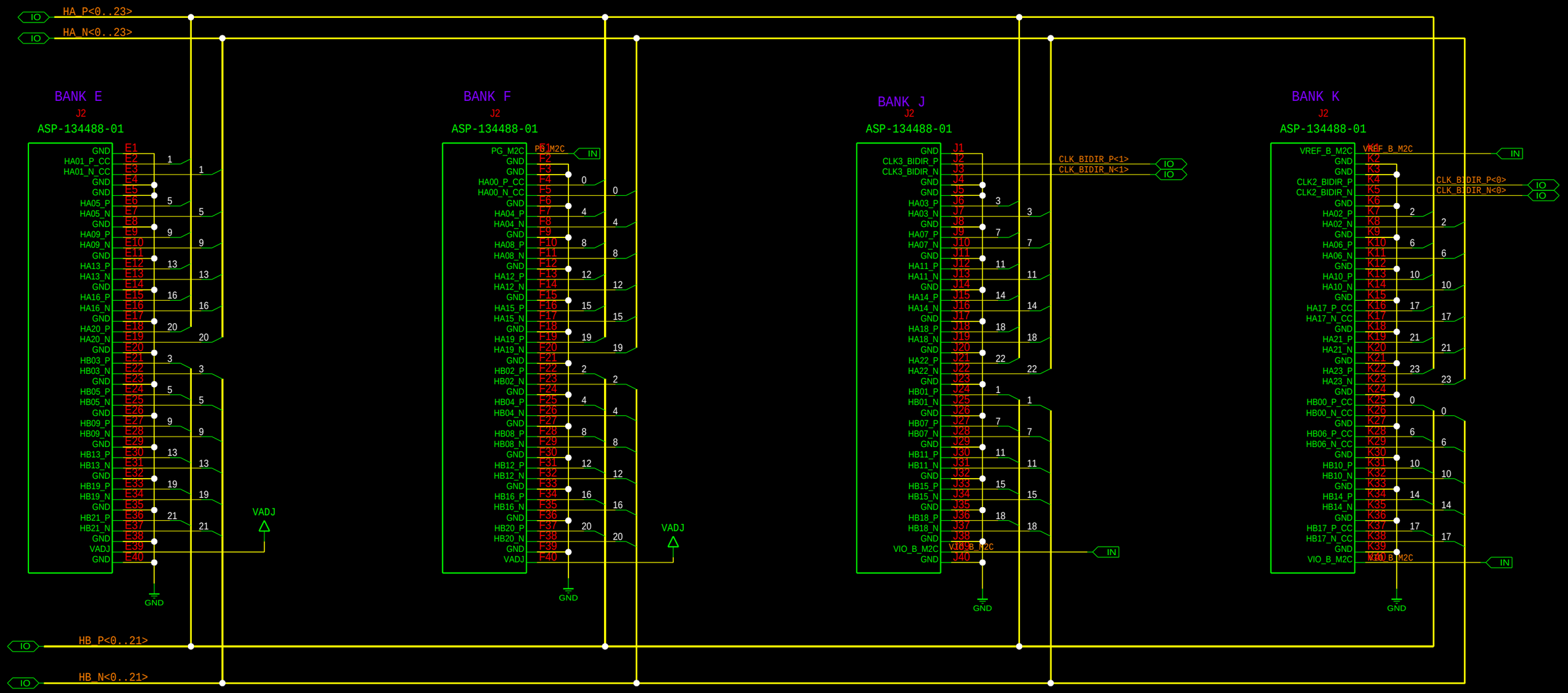

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PARTICLE PHYSICS

UNIVERSITY OF BRISTOL
H. H. WILLS PHYSICS LABORATORY
TYNDALL AVENUE, BS8 1TL
BRISTOL, UK

REVISION:	3	DATE MODIFIED:	2/March/2026
DESIGNED BY:	P. Baesso, D. Cussans, M. Alvarez	SHEET:	31 / 31

FMC HPC EFJK